

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

20EC502 – COMPUTER NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	List and explain the fundamental characteristics of data communication.	10	L*2	1	1
b)	Describe the OSI reference model in detail with a neat diagram.	10	L2	1	1
2. a)	For each of the following four networks, discuss the consequences if a connection fails.				
	i) Five devices arranged in a mesh topology				
	ii) Five devices arranged in a star topology (not counting the hub)				
	iii) Five devices arranged in a bus topology				
	iv) Five devices arranged in a ring topology	10	L3	1	1
b)	Draw and explain:				
	i) hybrid topology with a star backbone and three ring networks.				
	ii) hybrid topology with a ring backbone and two bus networks.	10	L2	1	1
Unit – II					
3. a)	Define framing and mention its classifications.	4	L2	2	1
b)	Describe bit stuffing and byte stuffing with examples.	8	L2	2	1
c)	Discuss with diagram SELECTIVE REPEAT ARQ flow control mechanism.	8	L2	2	1
4. a)	A code word 101001110001 is received at the receiver which uses a divisor 10111 . Check whether there is any error using CRC method.	4	L3	2	1
b)	Describe the Stop and Wait protocol with neat diagrams.	8	L2	2	1
c)	Explain HDLC frame format in detail.	8	L2	2	1
Unit – III					
5. a)	Describe the controlled access methods used in data communication in detail.	10	L2	3	1
b)	List and explain the methods used to reduce collision in CSMA with relevant diagrams.	10	L2	3	1
6. a)	Distinguish between FDMA, TDMA and CDMA channelization methods.	6	L3	3	1
b)	Describe the Ethernet standards and its evolution towards Full Duplex communication.	10	L2	3	1
c)	Compare Pure ALOHA and Sotted ALOHA.	4	L2	3	1
Unit – IV					
7. a)	Define the following:				
	i) Bridge				
	ii) Passive hub				
	iii) Repeater				
	iv) Backbone				
	v) Amplifier	10	L2	4	1
b)	Describe virtual LAN with an example. List the characteristics used to group stations in a virtual LAN and explain how the stations are configured in virtual LANs.	10	L3	4	1

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| 8. a) | Discuss network layer services in detail. | 10 | L2 | 4 | 1 |
| b) | An ISP is granted a block of addresses starting with 190.100.0.0/16(65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:
i) The first group has 64 customers; each need 256 addresses.
ii) The second group has 128 customers; each need 128 addresses.
iii) The third group has 128 customers; each need 64 addresses.
Design the subblocks and find out how many addresses are still available after these allocations. | 10 | L3 | 4 | 1 |

Unit – V

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|--------|--|----|----|---|---|
| 9. a) | Describe the forwarding process and forwarding techniques used by the network layer. | 10 | L2 | 5 | 1 |
| b) | Describe TCP header in detail with the relevant diagram. | 10 | L2 | 5 | 1 |
| 10. a) | List routing protocols and explain any one of them. | 10 | L3 | 5 | 1 |
| b) | What is meant by domain name System? Name different domain name spaces in application layer and explain them in brief. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
